

Project Phase 3 Report

**Reducing False Negatives in Breast Cancer Diagnosis Using a Hybrid SMOTE +
Cost-Sensitive XGBoost Framework**

ECOM 6306 – Supervised Machine Learning

Computer Engineering Department – IUG

Supervisor:

Dr. Ayman Maliha

Date of Submission:

December 2025

Group Members

Ashraf Suhail Ahmed Al-Kahlout (ID: 120250206)

Baraa Maher Shaaban AlDeeb (ID: 120232620)

Mahmoud Waleed Al-Sheikh Khalil (ID: 120231320)

Phase IV Report

Improving Breast Cancer Diagnosis by Reducing False Negatives Using ADASYN and Feature Selection

Introduction .1

In the previous phase of this project, a hybrid machine learning model combining SMOTE and XGBoost was implemented to address the class imbalance problem in breast cancer diagnosis. While this approach improved recall compared to the baseline model, further analysis revealed limitations related to synthetic noise and instability caused by traditional oversampling techniques. These limitations negatively affected the clinical reliability of the .model

Phase IV aims to improve the previously proposed solution by addressing its identified shortcomings. This phase introduces a refined hybrid learning framework that integrates feature selection, ADASYN oversampling, and cost-sensitive XGBoost. The objective is to further reduce false negatives while maintaining stable classification performance suitable for .medical decision support

Identified Shortcomings of Phase III Solution .2

Although the Phase III hybrid model achieved improved recall, it exhibited several important limitations. The use of SMOTE generated synthetic samples uniformly across the minority class feature space. In datasets with overlapping features between benign and malignant cases, this approach introduced synthetic noise that negatively affected the decision .boundaries learned by the classifier

Additionally, the Phase III model utilized all available features without evaluating their relative importance. This increased the risk of learning noise from less informative features, leading to reduced model stability and an increase in false positives. These issues highlighted the need for a more adaptive oversampling strategy and a dimensionality reduction .mechanism to improve clinical reliability

Proposed Improved Solution .3

To overcome the limitations observed in Phase III, Phase IV proposes an enhanced hybrid :framework consisting of three main components

Feature Selection: .1

Feature selection is applied using SelectKBest to retain only the most informative

features. This step reduces dimensionality, limits feature overlap, and prevents the model from learning noise from irrelevant attributes

ADASYN Oversampling: .2

ADASYN replaces SMOTE as the oversampling technique. Unlike SMOTE, ADASYN focuses on generating synthetic samples in regions of the feature space where classification is more difficult. This adaptive strategy reduces synthetic noise and improves the classifier's ability to learn complex decision boundaries

Cost-Sensitive XGBoost: .3

A cost-sensitive version of XGBoost is employed by assigning higher misclassification costs to malignant cases. This ensures that false negatives are penalized more heavily during training, further improving sensitivity toward cancer detection

Methodology .4

The Breast Cancer Wisconsin (Diagnostic) dataset was used in this phase. The dataset consists of numerical features extracted from digitized images of breast tissue biopsies and includes both benign and malignant samples. Malignant cases were treated as the positive class to prioritize clinical safety

The dataset was divided into training and testing sets using an 80%–20% stratified split to preserve class distribution. Feature scaling was applied using standardization. Feature selection was then performed on the training data before applying ADASYN oversampling. The final model was trained using the resampled dataset and evaluated on the untouched test set

Performance evaluation focused on recall, precision, F1-score, AUC-ROC, and the number of false negatives. These metrics provide a comprehensive assessment of both classification performance and clinical reliability

Experimental Results .5

The experimental results demonstrate that the Phase IV model achieved improved performance compared to the Phase III hybrid solution. The integration of feature selection reduced dimensional overlap, while ADASYN improved the quality of synthetic samples by focusing on difficult minority-class instances

Confusion matrix analysis shows a clear reduction in false negatives in the Phase IV model. ROC curve analysis further confirms improved class separability and model stability. Overall,

the proposed solution maintains high recall while reducing diagnostic risk associated with missed cancer cases

Comparative Analysis with Phase III .6

A direct comparison between Phase III and Phase IV highlights the effectiveness of the proposed improvements. While Phase III improved recall through SMOTE, it suffered from synthetic noise and instability. Phase IV successfully addresses these issues by combining adaptive oversampling and feature selection

The number of false negatives is reduced in Phase IV, indicating improved sensitivity toward malignant cases. At the same time, classification stability is enhanced, making the model more suitable for real-world clinical applications

Discussion .7

The results obtained in Phase IV demonstrate the importance of critically evaluating imbalance-handling techniques rather than applying them blindly. Although SMOTE is effective in many scenarios, its limitations become evident in medical datasets with overlapping features

The proposed Phase IV framework provides a more balanced and clinically reliable solution by focusing on hard-to-classify cases and reducing noise. This approach aligns well with the primary objective of medical diagnosis, which prioritizes patient safety over purely statistical performance

Conclusion .8

Phase IV of this project presented an improved hybrid learning framework for breast cancer diagnosis. By integrating feature selection, ADASYN oversampling, and cost-sensitive XGBoost, the proposed solution effectively addressed the shortcomings identified in Phase III

The final model achieved a lower number of false negatives while maintaining stable overall performance. These results demonstrate that adaptive oversampling and dimensionality reduction play a crucial role in improving the clinical reliability of machine learning-based diagnostic systems

Future Work .9

Future research may explore advanced feature selection techniques, hyperparameter optimization, and deep learning models to further improve diagnostic performance. Additionally, evaluating the proposed framework on larger and more diverse medical datasets

- would enhance its generalizability and clinical applicability